UNIVERSITY OF VAVUNIYA

Faculty of Technology Studies

Department of ICT

SYSTEM ANALYSIS AND DESIGN
INDIVIDUAL ASSIGNMENT
PROFESSIONAL PORTFOLIO WEBSITE

.

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Index Number: 2022/ICTS/07

Course Code: TICT2252

Degree Program: BICT Honours

Date of Submission: 12/08/2026

Website URL:https://chathuri-nv.github.io/Professional-Portfolio/

GitHub Repository URL: https://github.com/chathuri-nv/Professional-Portfolio/tree/main

1. Problem Statement & Objectives

Problem Statement :

Many students have skills, academic achievements and project experience, but they do not have a professional online platform to present them in an organized and attractive way. Sharing information through documents or social media is not effective for showcasing personal achievements. Therefore, a professional portfolio website is needed to present personal information, skills, education, projects and contact details in one place.

Objectives :

- ✓ Create a professional, mobile-responsive portfolio website.
- ✓ Present education, skills, and projects clearly.
- ✓ Provide Downloadaemployers and lecturers with easy access to the CV and contact information.
- ✓ Include links to LinkedIn and GitHub.
- ✓ Deploy the site online for free access.

2. Stakeholders, Scope & Constraints

Stakeholders:

- **Student (Developer):** Responsible for designing and building the portfolio website.
- **Employers:** Intended audience who will review the portfolio to evaluate skills and projects.
- **Lecturers:** Assess the assignment for academic grading and feedback.

Scope:

- The project focuses on a **front-end portfolio website** using HTML, CSS, Bootstrap, and JavaScript.
- No backend database or server-side programming is included.
- Hosting will be done on **free platforms** such as GitHub Pages, Netlify, or Verel.
- The website will include sections for education, skills, projects, CV, and external links.

Constraints

- **Time:** Must be completed within 2–3 weeks.
- **Resources:** Limited to beginner-friendly tools and free hosting services.
- **Quality:** Must remain professional, accurate, and visually appealing.
- **Accessibility:** Should be responsive and easy to navigate on both desktop and mobile devices.

3. Requirements

Functional Requirements

- The website must allow navigation between sections (Home, About, Projects, Contact).
- Users should be able to view and download the CV.
- Projects must be displayed with descriptions and screenshots.
- External links to LinkedIn and GitHub should be accessible.
- A contact option (email link or form) should be available.

Non-Functional Requirements

- The website must be **responsive** (work on desktop, tablet, and mobile).
- It should have a **professional and clean design**.
- Pages should load quickly with optimized images.
- The site must be **easy to update** in the future (simple HTML/CSS edits).
- Accessibility should be considered (clear fonts, readable colors, alt text for images).

4. Use-case Diagram & Descriptions

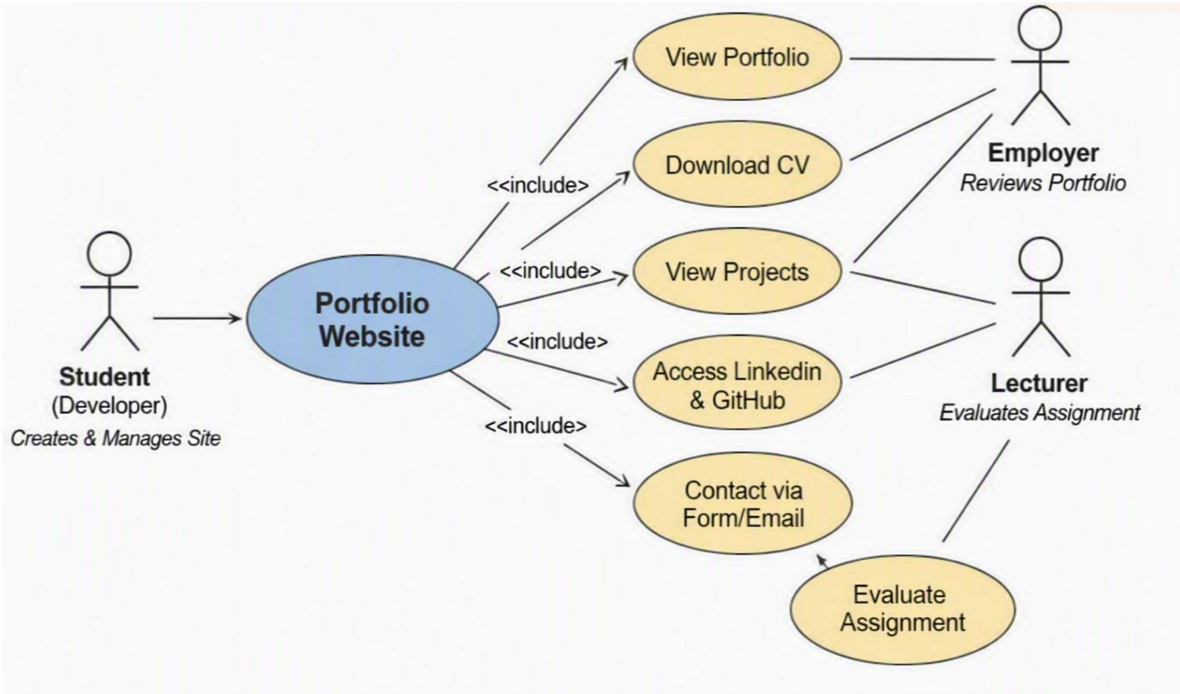

Figure 1: Use Case Diagram for Portfolio Website

Description of Use Cases

Actors:

- **Student (Developer):** Creates and manages the portfolio website.
- **Employer:** Views portfolio content, downloads CV, checks projects.
- **Lecturer:** Reviews the portfolio for assignment evaluation.

Main Use Cases:

1. **View Portfolio:** Employers and lecturers can browse the website.
2. **Download CV:** Employers can download the student’s CV.
3. **View Projects:** Employers can check project details and screenshots.
4. **Access External Links:** Employers can visit LinkedIn and GitHub profiles.
5. **Evaluate Assignment:** Lecturers assess the portfolio against course requirements.

5. Activity Diagram

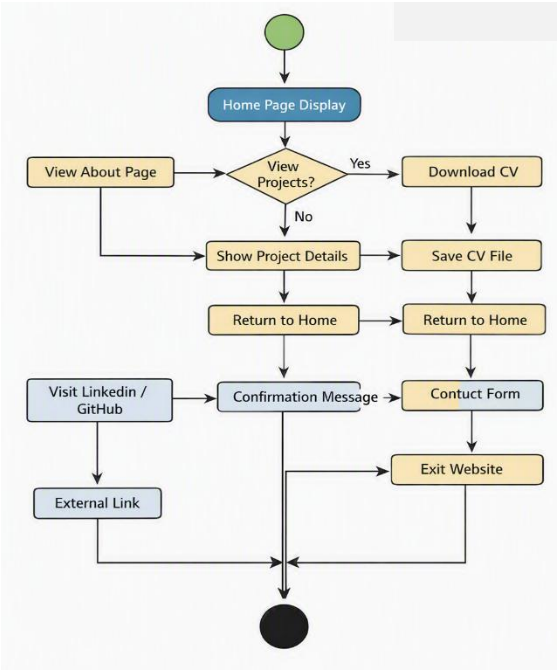

Figure 2: Activity Diagram for Portfolio Website

Description ::

The activity diagram illustrates the sequence of actions a user performs when interacting with the portfolio website. It begins with the **Start** node, followed by the **Home Page Display**. From there, users can navigate to different sections such as **About**, **Projects**, **CV Doswnload**, and **Contact Form**. Each activity leads to a decision point — for example, whether the user wants to view projects or download the CV. The process ends at the **Exit Website** node once the user finishes browsing.

Key Activities::

- **Start → Home Page Display**
- **Navigate to About Section**
- **View Projects → Project Details → Return to Home**
- **Download CV → Save File → Return to Home**
- **Access LinkedIn/GitHub → External Redirect**
- **Contact Form → Submit → Confirmation Message**
- **End → Exit Websit**

6. Site Map & Wireframes

Site Map

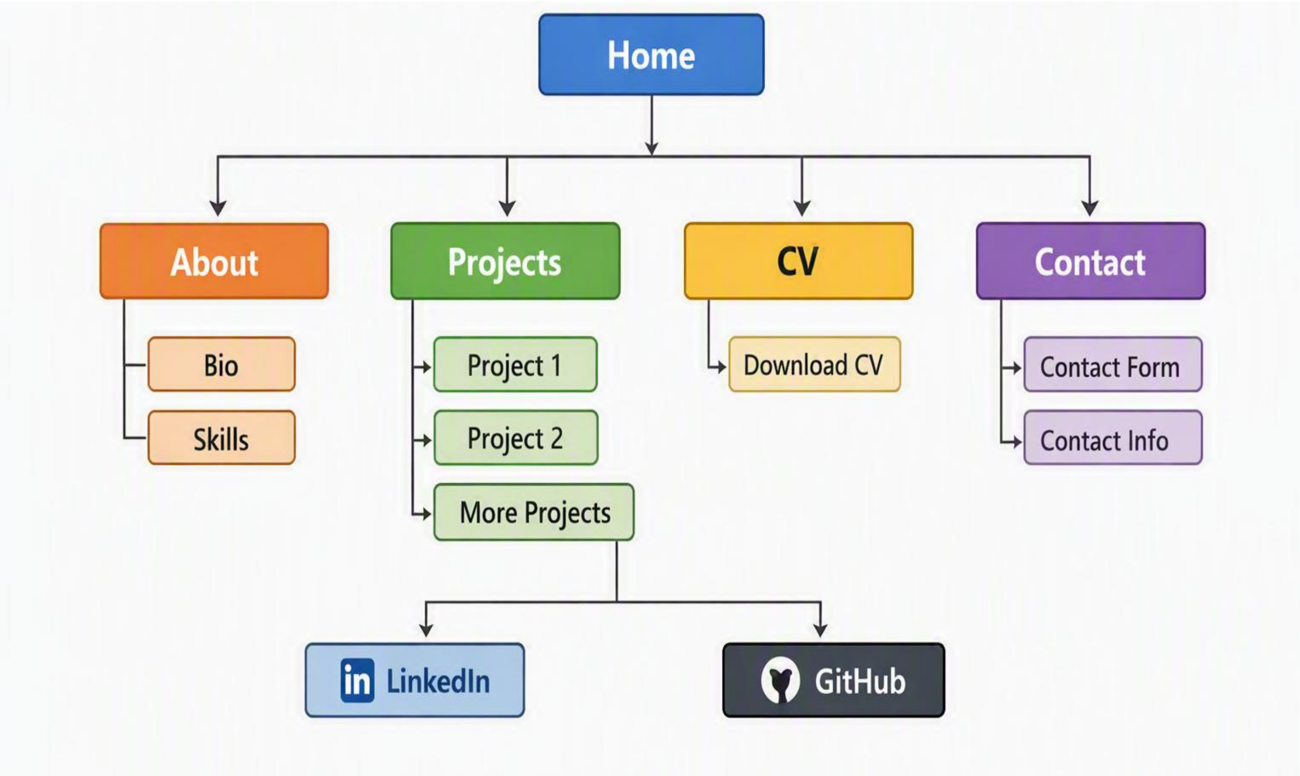

Figure 3:Site Map for Portfolio Website

Description :

The site map outlines the overall structure of the portfolio website. It begins with the **Home Page**, which links to the **About**, **Projects**, **CV**, and **Contact** sections. Each section provides specific information — for example, the **Projects** page displays project details and screenshots, while the **CV** page allows users to download the student's résumé. External links to **LinkedIn** and **GitHub** are accessible from the footer or navigation bar.

Wireframes

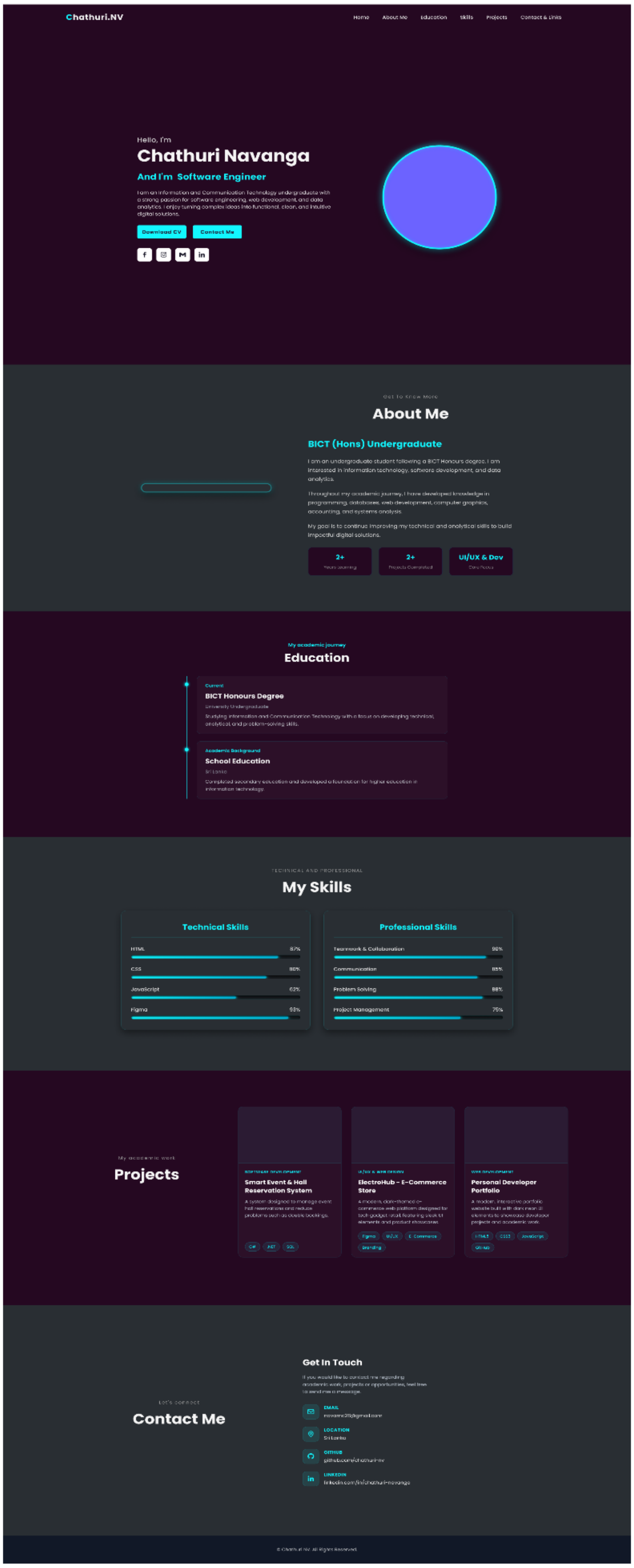

Figure 4:My Wire Frame

7. Implementation Summary

Development Process

The portfolio website was implemented using **HTML, CSS, Bootstrap, and JavaScript**. The design followed the wireframes created earlier, ensuring a clean and professional layout. Bootstrap was used for responsiveness, while custom CSS provided styling consistency. JavaScript was applied for interactive elements such as navigation and form validation.

Hosting & Deployment

- The website was deployed on **GitHub Pages** for free hosting.
- Version control was maintained using **GitHub repositories**.
- External links to **LinkedIn** and **GitHub** were tested to ensure accessibility.

Testing & Validation

- The site was tested on multiple devices (desktop, tablet, mobile) to confirm responsiveness.
- All links and buttons were verified for functionality.
- The CV download feature was tested to ensure the file is accessible.
- Accessibility checks included font readability, color contrast, and alt text for images.

Challenges & Solutions

- **Challenge:** Limited time (2–3 weeks).

  Solution: Focused on essential features first, then added enhancements.

- **Challenge:** Ensuring responsiveness.

  Solution: Used Bootstrap grid system and media queries.

- **Challenge:** Budget constraints.

  Solution: Used free hosting and open-source tools.

8. Testing Table

Test Case ID	Feature Tested	Input/Action	Expected Output	Actual Output	Status
TC-01	Home Page Load	Open website URL	Home page displays correctly	Home page displays correctly	Pass
TC-02	Navigation Links	Click “About” link	About page opens	About page opens	Pass
TC-03	Projects Section	Click “View Details” button	Project details page opens	Project details page opens	Pass
TC-04	CV Download	Click “Download CV” button	CV file downloads successfully	CV file downloads successfully	Pass
TC-05	Contact Form	Enter valid data → Submit	Confirmation message displayed	Confirmation message displayed	Pass
TC-06	Contact Form	Leave fields empty → Submit	Error message displayed	Error message displayed	Pass
TC-07	External Links	Click LinkedIn/GitHub icons	Redirects to external profile	Redirects to external profile	Pass
TC-08	Responsiveness	Open site on mobile device	Layout adjusts to screen size	Layout adjusts to screen size	Pass

9. Reflection on Career Value & SAD Learning

Reflection on Career Value

Developing the portfolio website provided significant career value. It created a professional platform to showcase academic achievements, technical skills, and projects to potential employers. The project also strengthened confidence in presenting work visually and technically, which is essential for career growth in ICT. By practicing real-world implementation, the student gained practical experience that can be highlighted in interviews and used to build a stronger professional identity.

Reflection on SAD Learning

The project reinforced key concepts from Systems Analysis & Design (SAD). Creating use-case diagrams, activity diagrams, site maps, and wireframes helped translate theoretical knowledge into practical application. The process of identifying stakeholders, defining requirements, and testing functionality demonstrated the importance of structured analysis before implementation. This experience highlighted how SAD principles ensure clarity, reduce errors, and improve project outcomes.

Overall, the project bridged classroom learning with practical execution, showing how SAD methodologies directly support successful system development.

10. LinkedIn Evidence

As part of the career-oriented value of this project, the portfolio website was connected to the student's **LinkedIn profile**. This integration ensures that employers and lecturers can verify professional details and view additional achievements. The LinkedIn profile serves as external evidence of the student's skills, certifications, and projects.

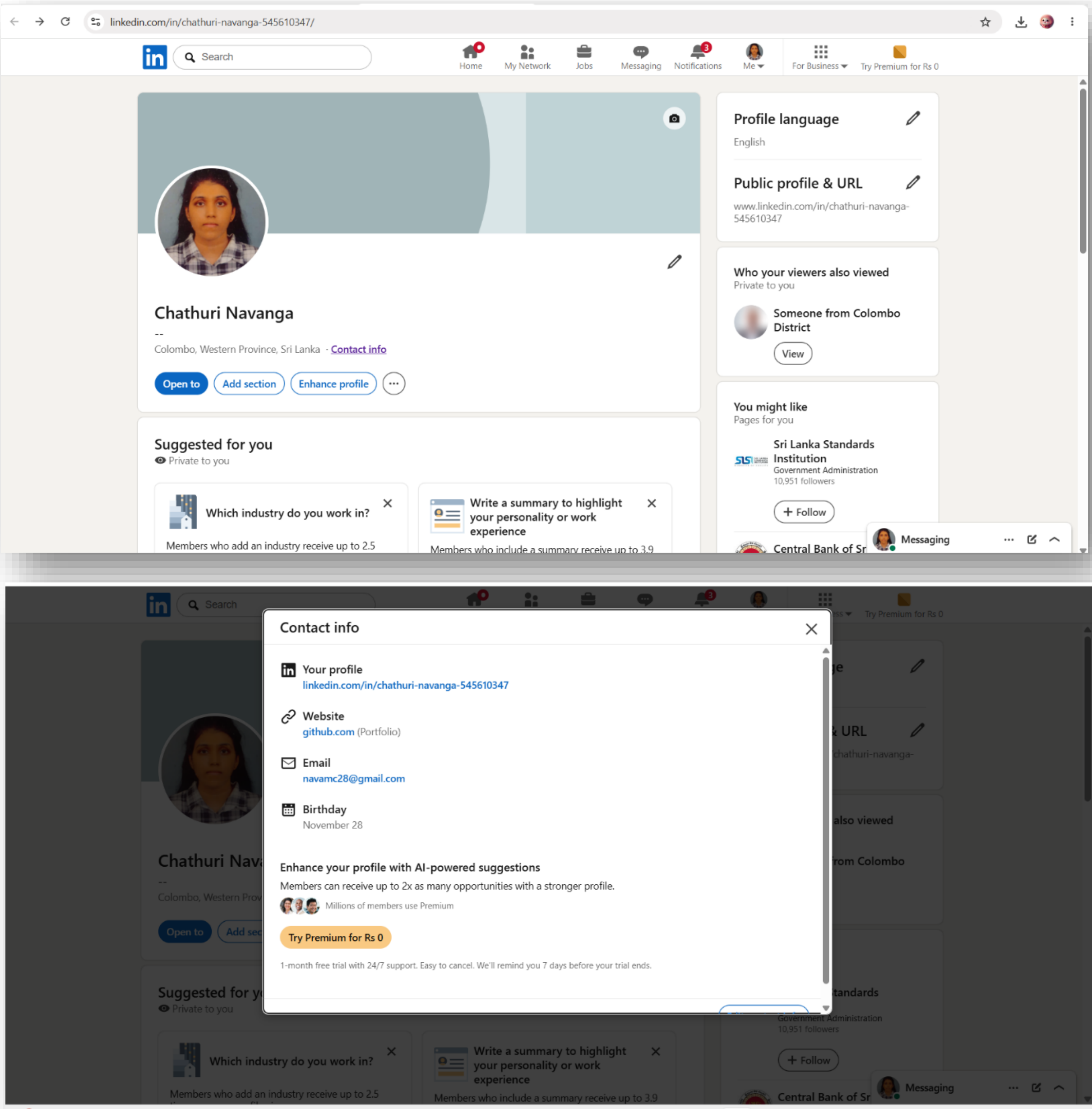

Figure 5: LinkedIn Profile Evidence

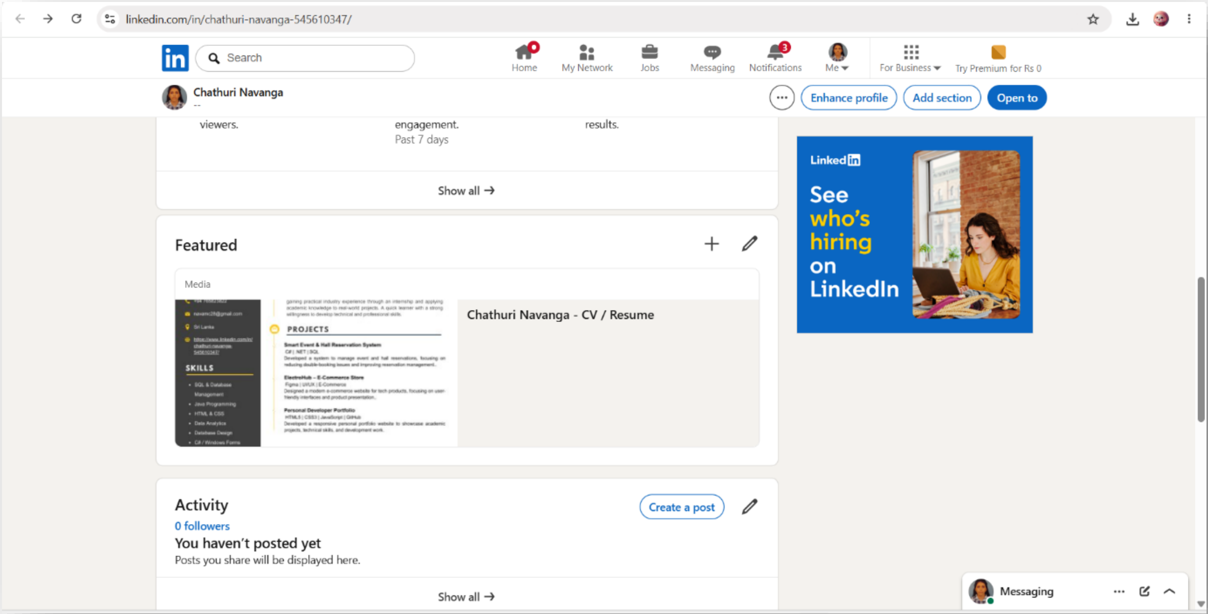

Figure 6:LinkedIn Projects Evidence

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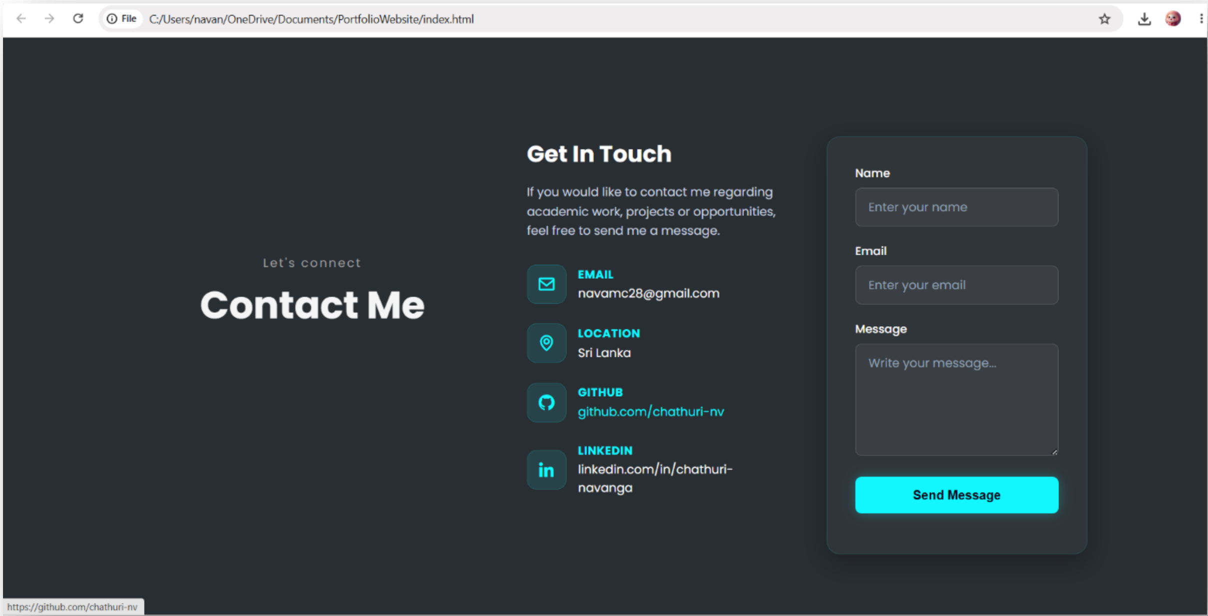

Figure 7: LinkedIn Connect to HTML Evidence